

ELEC5305 Project Proposal

1. Project Title

Environmental Sound Classification Using MFCC, Mel Spectrograms, and CNNs

2. Student Information

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3. Project Overview

This project investigates environmental sound classification: given a short audio recording, the system predicts the type of sound event that it contains. The selected task uses the ten classes in the UrbanSound8K dataset, including air conditioner, car horn, children playing, dog bark, drilling, engine idling, gun shot, jackhammer, siren, and street music. Unlike speech recognition, the objective is to recognise non-speech acoustic events that may vary greatly in duration, loudness, recording conditions, and background noise.

The proposed work will compare two audio representations and classification approaches. The baseline system will summarise each recording with Mel-Frequency Cepstral Coefficients (MFCCs) and classify the resulting feature vector using a compact multilayer perceptron (MLP). The improved system will convert audio into log-Mel spectrograms and use a two-dimensional convolutional neural network (2D CNN) to learn local time-frequency patterns. Data augmentation and regularisation will then be evaluated as additional improvements.

The project is closely related to the course topics of framing, windowing, the Short-Time Fourier Transform (STFT), spectral representation, Mel filter banks, MFCCs, and audio-feature design. The final system will provide reproducible experiments, quantitative comparisons, confusion-matrix visualisation, and example predictions for common urban sounds.

4. Background and Motivation

Environmental sound classification supports applications such as smart-city monitoring, public safety, context-aware mobile devices, domestic robots, and assistive systems. The task is challenging because recordings from the same class can sound different, while acoustically similar classes may overlap. For example, drilling and jackhammer sounds may share strong repetitive spectral patterns. A useful classifier must therefore capture both the spectral content and its evolution over time.

This topic was selected for three reasons. First, it strongly aligns with ELEC5305 because both MFCCs and Mel spectrograms are constructed through framing, windowing, FFT-based spectral analysis, Mel filter banks, and logarithmic

compression. Second, it enables a clear baseline-to-improvement comparison: MFCC summary features provide a compact conventional representation, while a 2D CNN can learn patterns directly from a time-frequency image. Third, UrbanSound8K provides an established ten-class dataset and official folds, making the evaluation reproducible and reducing the risk of data leakage.

The main research question is: How do MFCC and log-Mel spectrogram representations affect the accuracy and robustness of environmental sound classification, and how much further improvement can be obtained through CNN regularisation and audio augmentation? The project hypothesises that the Mel-spectrogram CNN will outperform the MFCC-MLP baseline because it preserves temporal structure, while augmentation will improve generalisation to unseen recording conditions.

5. Proposed Methodology

5.1 Dataset and Experimental Protocol

- Use UrbanSound8K, which contains 8,732 labelled audio excerpts of up to four seconds across ten urban sound classes.
- Resample recordings to mono at 22.05 kHz, normalise amplitude, and pad or crop each example to a fixed four-second duration. The same preprocessing rules will be used for all models.
- Respect the official predefined folds. Folds 1-8 will be used for training, fold 9 for validation, and fold 10 for final testing. No recordings from the test fold will be used for model selection.
- Report the class distribution and use class weights or balanced sampling if class imbalance materially affects training.

5.2 Audio Representations

- Apply short-time analysis using a Hann window, an FFT size of 1,024 samples, and a hop length of 512 samples. These parameters will be held constant unless a controlled parameter experiment is conducted.
- MFCC representation: compute approximately 40 MFCCs per frame and aggregate them using statistics such as mean and standard deviation to obtain a fixed-length feature vector for the baseline classifier.
- Mel-spectrogram representation: compute a 128-band log-Mel spectrogram and standardise the input before passing it to the CNN.
- Visualise waveforms, STFT spectrograms, MFCCs, and Mel spectrograms for representative classes to explain the information preserved by each representation.

5.3 Classification Models

- Baseline model: train a small MLP using the aggregated MFCC feature vector. It will contain one or two hidden layers, ReLU activation, and a ten-class softmax output.
- Improved model: train a 2D CNN on log-Mel spectrograms using convolution, batch normalisation, ReLU activation, max pooling, dropout, and a final softmax classifier.

- Use cross-entropy loss and Adam optimisation. Validation loss will guide early stopping and learning-rate adjustment. Fixed random seeds and saved configurations will support reproducibility.

5.4 Experiments and Evaluation

- Experiment A - Establish baseline performance using aggregated MFCC features and an MLP.
- Experiment B - Train a 2D CNN on log-Mel spectrograms and compare it with the baseline under the same data split.
- Experiment C - Add waveform augmentation, including time shifting, random gain, background-noise mixing, and optional pitch or time-stretch transformations.
- Experiment D - Conduct ablation tests for dropout, batch normalisation, and augmentation to determine which changes contribute to performance.
- Experiment E - Test robustness using additional noise at 20 dB, 10 dB, and 0 dB SNR and compare the degradation of the baseline and improved systems.
- Report overall accuracy, macro-averaged precision/recall/F1-score, per-class recall, confusion matrices, training time, model size, and average inference time. Where feasible, repeated runs will be used to report mean performance and variability.

5.5 Implementation Tools

- Python will be used with Librosa/SciPy for audio processing, NumPy and pandas for data handling, scikit-learn for the baseline and metrics, PyTorch or TensorFlow for CNN training, and Matplotlib/Seaborn for visualisation.
- The GitHub repository will contain modular source code, configuration files, reproducible training and evaluation scripts, a README, selected result figures, and instructions for obtaining UrbanSound8K. The dataset itself will not be committed.

6. Expected Outcomes

1. A complete environmental-sound classification pipeline covering audio preprocessing, MFCC and Mel-spectrogram extraction, model training, testing, and evaluation.
2. A reproducible MFCC-MLP baseline and a clearly documented log-Mel-spectrogram CNN improvement.
3. Tables and figures comparing accuracy, macro-F1, noise robustness, model size, and computational cost across experimental conditions.
4. Confusion matrices and representative time-frequency plots that explain common class confusions and the effect of the selected audio representation.
5. A discussion of limitations, including class imbalance, recording variability, dataset size, background noise, and generalisation beyond UrbanSound8K.
6. A GitHub project page containing code, documentation, result visualisations, and a short demonstration. The final deliverables will also include the course report and video presentation required by the unit.

7. Timeline (Weeks 6-13)

Week	Task
6-7	Literature review; obtain UrbanSound8K; inspect labels and prepare the official-fold data pipeline.
8	Implement preprocessing, MFCC extraction, and log-Mel-spectrogram visualisation.
9	Implement and evaluate the MFCC-MLP baseline classifier.
10	Implement and tune the 2D CNN using log-Mel spectrograms.
11	Conduct augmentation, ablation, and noise-robustness experiments.
12	Analyse metrics and confusion matrices; organise code, README, and GitHub project page.
13	Complete the final report, code demonstration, and video presentation.

8. References

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